L6 home assignment — Quantization + BF16 vs on-the-fly 8-bit serving. # Use a FRESH virtualenv. Install on your GPU box (H100 or L40S) for Q2. # # vLLM pins its own compatible torch build, so we do NOT pin torch separately # (pinning both is the usual source of CUDA/torch conflicts). Install vLLM first, # let it bring torch, then the rest. # Q2's quantized config is GPU-adaptive: `--quantization fp8` on an H100 (needs Hopper), # `--quantization int8_per_channel_weight_only` on an L40S / non-FP8 GPU. The int8 method is # recent, so use a current vLLM. If your build rejects it, `vllm serve --help | grep -i # quantization` lists what it supports (see INSTRUCTOR.md for the W8A8-checkpoint fallback). vllm>=0.24.0 # latest; older releases may not have int8_per_channel_weight_only guidellm==0.7.1 # `guidellm run` CLI (0.7.x). 0.4.x breaks against the transformers vllm pulls. matplotlib==3.9.2 numpy>=1.26 jupyterlab==4.2.5 ipykernel==6.29.5 # Q1 is pure PyTorch and runs anywhere (CPU is fine for the self-check). # Q2 requires a GPU (H100 or L40S). If you only want to smoke-test Q1 without vLLM: # pip install torch matplotlib jupyterlab ipykernel